

OSPO: Object-Centric Self-Improving Preference Optimization for Text-to-Image Generation

Yoonjin Oh, Yongjin Kim, Hyomin Kim, Donghwan Chi, Sungwoong Kim*
Korea University

{dhdbsr1w, rla020, khmjee, zheedong, swkim01}@korea.ac.kr

Abstract

Recent advances in Multimodal Large Language Models (MLLMs) have enabled unified multimodal understanding and generation. However, they still struggle with fine-grained text–image alignment, often failing to faithfully depict objects with correct attributes such as color, shape, and spatial relations. To mitigate this issue, previous studies have explored preference optimization methods such as DPO and GRPO, but these approaches incur substantial computational cost, both in constructing preference data and in performing optimization. This has motivated self-improving preference optimization approaches, in which the MLLM autonomously generates its own training data, self-estimates preference feedback, and self-optimizes using the resulting self-constructed preference pairs. However, existing self-improving methods still overlook fine-grained, object-level semantics, allowing object hallucination to persist. To tackle this problem, we propose Object-centric Self-improving Preference Optimization (OSPO), a self-improving framework designed to enhance object-level text–image alignment. OSPO explicitly constructs object-centric preference data without relying on any external data and external models. We also introduce a new approach that leverages attention-based object masks together with an object-weighted SimPO loss to enhance object-specific fidelity. Extensive experiments on three compositional image generation benchmarks demonstrate that OSPO significantly improves fine-grained alignment and reduces object hallucination, outperforming prior self-improving methods and even specialized diffusion-based text-to-image models.

1. Introduction

Recent advancements in Unified Multimodal Large Language Models (Unified MLLMs) [8, 29–31, 35, 38–40] have achieved remarkable performance, demonstrating the ability to perform diverse vision tasks ranging from image understanding to generation within a single model.

Despite these successes, a persistent challenge remains in achieving fine-grained alignment between text and image. In text-to-image (T2I) generation, MLLMs often fail to faithfully render the details specified in the input prompt, including spatial relationships and precise visual attributes of objects. A critical and related failure mode is object hallucination, which encompasses not only the generation of non-existent objects but also the omission or distortion of objects described in the prompt.

Early work has addressed fine-grained alignment and hallucination using feedback-based post-training methods such as PPO [27] and DPO [26, 34]. Despite their effectiveness, these approaches require large amounts of human- or AI-curated preference data, and collecting such data for image generation is far more complex and costly than for text generation, posing serious scalability challenges. Moreover, these methods inherently suffer from off-policiness. PPO relies on reward models trained on externally sourced preference data, and DPO depends on preference pairs produced by humans or stronger models. These factors create distributional mismatches between the external preference data and the model’s own output distribution, leading to unstable optimization and further limiting the scalability of fine-grained alignment.

To overcome these constraints, recent works have introduced self-improving approaches [11, 20, 24] that eliminate reliance on external data or models by generating training data or reward signals internally. This direction is especially promising with the advent of Unified MLLMs, whose dual abilities in image understanding and generation enable the model to evaluate its own outputs and form a self-improving loop. Furthermore, because MLLMs’ image understanding capability has improved rapidly, their visual comprehension is generally more reliable than their generative capability. This stronger understanding ability allows the model to effectively guide and improve its own image generation within the self-improving loop. However, despite these strengths, current self-improving methods still lack mechanisms that explicitly enforce fine-grained text–image alignment, leaving them insufficient for fully resolving object-level hallucination and detailed alignment failures.

In this paper, we propose Object-centric Self-improving

*Corresponding author

Preference Optimization (OSPO), a self-improving framework that autonomously constructs fine-grained, object-focused T2I preference pairs and performs object-centric preference optimization guided by object masks. This design enables the model to internalize object-specific knowledge and better capture fine-grained text–image correspondences. Specifically, OSPO operates through five stages. First, it generates initial text prompts categorized into four semantic types. Second, it perturbs each text prompt into multiple variations and enriches the original prompt with each perturbed one so that the two prompts in each pair share a similar global context but differ in fine-grained details. Third, it generates images from the enriched texts as candidate preference pairs and predicts an object mask for each image based on attention weights from intermediate layers. Fourth, it performs Self-VQA using self-generated decompositional questions, filters out incorrect image samples, and selects the image with the highest VQA score as the preferred one to form the final preference pair. Finally, it performs object-centric preference optimization using an object-weighted loss derived from the predicted object masks. We validate the effectiveness of OSPO across multiple MLLMs on three established benchmarks for fine-grained T2I generation. Notably, OSPO substantially improves fine-grained alignment, outperforming existing self-improving methods and even surpassing diffusion-based models specifically designed solely for image generation.

To summarize, our main contributions are as follows:

- We present OSPO, a five-stage self-improving framework that mitigates object hallucination in T2I generation without relying on either external datasets or auxiliary models.
- We propose a pipeline that constructs high-quality, object-centric preference data with shared global semantics but fine-grained local differences, and incorporates object-aware supervision using attention-based object masks with an object-weighted SimPO loss.
- We demonstrate the empirical effectiveness of OSPO with substantial object-level alignment improvements across multiple T2I benchmarks, surpassing prior self-improving and specialized diffusion models.

2. Related Works

2.1. Multimodal Large Language Models (MLLMs)

Early Multimodal Large Language Models (MLLMs), particularly large vision-language models [4, 6, 19, 43], primarily extended pretrained LLMs for visual understanding by integrating vision encoders. Recent advances, however, have moved toward unified MLLMs that can both understand and generate visual content. This unification enables a single model to handle diverse tasks, ranging from image understanding to image generation, within a shared set of model parameters. Among them, Discrete MLLMs [3, 8, 29, 30, 35, 37, 38] represent visual information as quantized tokens produced by vector quantization (VQ)

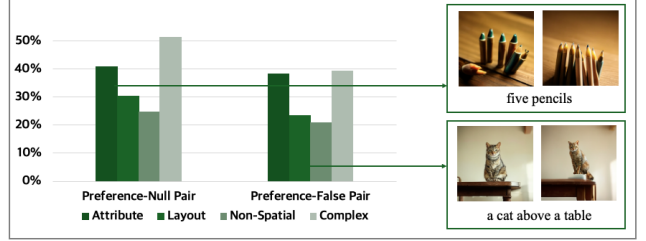


Figure 1. Examples and category distribution of (Left) *preference-null* and (Right) *preference-false* image pairs.

[7, 33] or learned tokenizers. These models employ vector quantization to discretize images into tokens, thereby allowing both vision and language to be modeled uniformly through next-token prediction. In this work, we also focus on discrete unified MLLMs that generate images as sequences of discrete tokens.

2.2. Self-Improving Frameworks for T2I

For T2I generation, SILMM [24] introduces the first self-improving PO framework for MLLMs. It adopts a *Best-of-N sampling* strategy to generate candidate images and assigns preference feedback using Self-VQA, similar to RLAIIF-V [44]. SILMM primarily targets improvements in compositional T2I generation. However, this objective is applied only during evaluation and is not reflected in its generation or training stages. Subsequent frameworks [11, 20] combine T2I generation with image understanding tasks but continue to rely on the same *Best-of-N sampling* paradigm. As they focus on general T2I quality rather than compositional or fine-grained image generation, they do not incorporate a component for object-level alignment in any stage of the framework.

Overall, existing self-improving frameworks lack fine-grained or object-centric design across their generation, evaluation, and optimization processes, limiting their ability to address object hallucination. In contrast, our framework explicitly integrates object-level, fine-grained components throughout all stages, enabling more precise and semantically faithful T2I generation.

3. Framework

3.1. Preliminary

We analyze the 16,000-sample self-generated preference dataset produced using Janus-Pro-7B [3] by following the workflow provided in SILMM [24]. We identify two systematic error patterns in the preferred and non-preferred image pairs: (1) **Preference-Null Pairs**, where both images exhibit similar levels of prompt fidelity, and (2) **Preference-False Pairs**, where a prompt-aligned image is incorrectly labeled as the non-preferred one. Both cases introduce noisy or contradictory supervision.

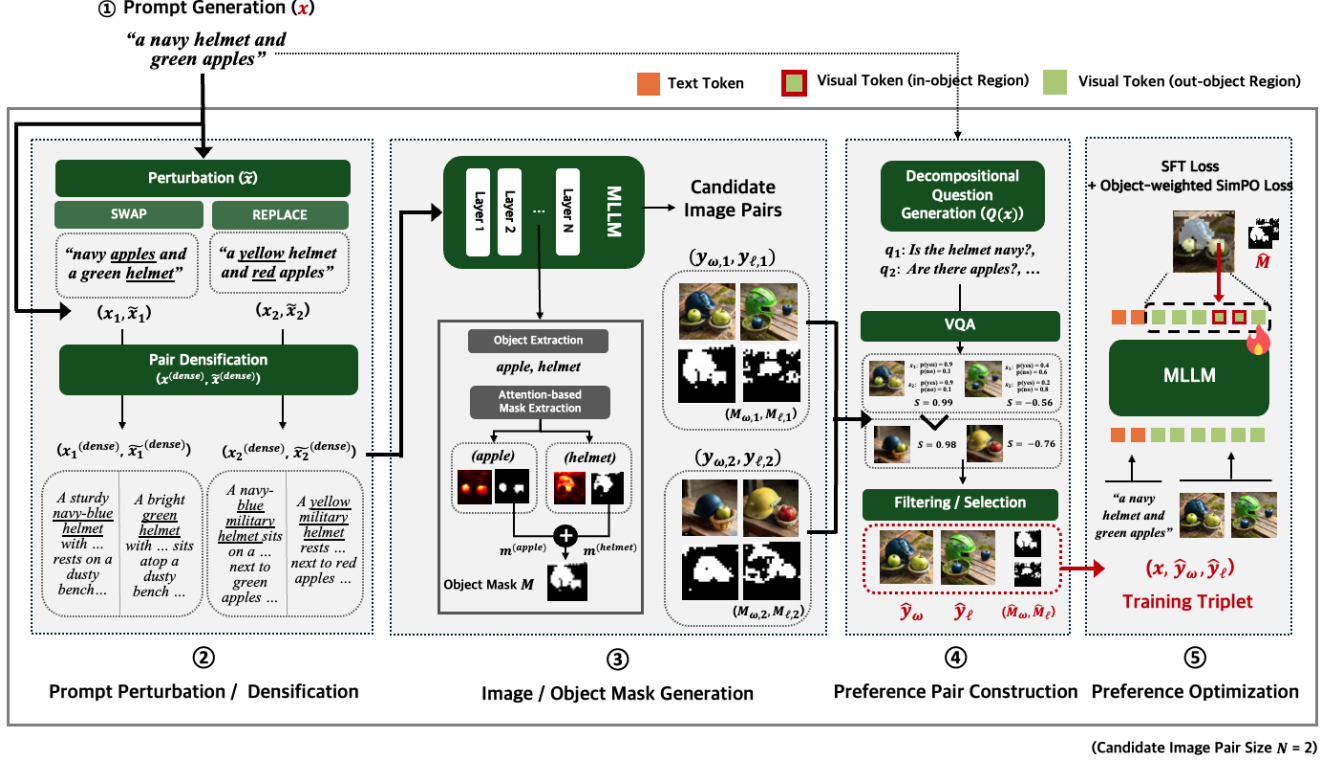


Figure 2. Overview of OSPO framework: (Stage 1) The MLLM generates a set of base text prompts. (Stage 2) For each base prompt, the model generates multiple perturbed variants using three strategies (Replace, Swap, and Drop) and each original-perturbed pair is jointly densified. Each pair of original and perturbed prompt is pairwise densified by the MLLM. (Stage 3) The MLLM generates candidate preferred and non-preferred images from each densified prompt pair. (Stage 4) The model constructs atomic decompositional VQA questions and evaluates each candidate image’s prompt fidelity using the Self-VQA alignment score S , filtering out noisy supervision and selecting a single final image pair. (Stage 5) The MLLM is fine-tuned using preference optimization with the proposed Object-weighted SimPO loss together with SFT losses.

As shown in Figure 1, a substantial portion of generated pairs across categories exhibits ambiguous or incorrect preference labels, making them effectively indistinguishable and providing little informative training signal. This arises from the nature of the *Best-of-N sampling* strategy, which is indiscriminate and driven by random sampling. Consequently, the dataset becomes noisy and poorly aligned with the training objective, which requires object-level corrective signals. The detailed experimental settings for this analysis are provided in the Appendix.

3.2. Overview

We introduce **OSPO**, a five-stage self-improving preference optimization framework designed to mitigate object hallucination in T2I generation. The framework is fully self-contained, relying neither on external data nor pretrained reward models. By leveraging the dual capabilities of unified MLLMs, OSPO synthesizes text-image data from scratch without relying on *Best-of-N sampling*. It then constructs object-centric preference pairs that deliver precise, fine-grained supervision, addressing limitations of prior self-

improving frameworks and mitigating object-level hallucination. The five stages are as follows: (1) Prompt Generation (2) Prompt Perturbation and Densification (3) Image and Object Mask Generation (4) VQA-based Preference Pair Construction (5) Preference Optimization. Figure 2 describes an overview of the OSPO framework.

3.3. Prompt Generation

Inspired by T2I-Compbench++ [15], we first construct the base set of training prompts, where each training prompt x serves as the basis for subsequent data construction. The prompt set consists of four categories: Attribute, Layout, Non-spatial Relationship, and Complex Composition. The Attribute category is decomposed into Color, Shape, and Texture, while Layout is decomposed into 2D and 3D-Spatial Relationships. The remaining categories are single-level categories. Non-spatial Relationship describes dynamic actions or states, while Complex Composition combines multiple categories to form richer and longer prompts. All prompts are generated by MLLMs via in-context learning. The prompt template is provided in the Appendix.

3.4. Prompt Perturbation and Densification

To construct a preferred and non-preferred image pair (y_w, y_ℓ) for each text prompt x , we generate the two images independently rather than using the common *Best-of- N sampling* strategy, which produces multiple candidates from the same prompt and selects the highest- and lowest-scoring ones. We therefore generate preferred and non-preferred image variants using prompt pairs that share the same global semantics but differ slightly in object-level details, while leveraging the MLLM’s in-context learning capability. The templates are presented in the Appendix.

3.4.1. Prompt Perturbation

Inspired by SugarCrep [12] and WinoGround [32], multiple perturbed prompts $\tilde{x}$ are generated for each text prompt x using the following perturbation strategies:

- **Replace**: substitutes an object or attribute in the prompt with another that was not originally present, creating a novel compositional combination.
- **Swap**: exchanges the positions of objects or attributes within the prompt, inducing a different relational binding from the original.
- **Drop**: removes an object or attribute from the prompt, introducing semantic ambiguity.

After generating N perturbed prompts, each perturbed prompt is paired with the original text prompt to form N distinct pairs: $(x, \tilde{x}_1), (x, \tilde{x}_2), \dots, (x, \tilde{x}_N)$. The original prompt x is shared across all pairs, while each $\tilde{x}_n$, $n \in \{1, \dots, N\}$ represents a different perturbation. In our experiments, we set $N = 3$ by default.

3.4.2. Prompt Densification

Prior work [18, 36, 41, 42] shows that *Prompt Densification*—adding contextual detail to prompts—improves faithfulness in T2I generation. We extend this idea by applying densification to each prompt pair $(x, \tilde{x})$ before image synthesis, ensuring that paired images share global semantics but differ only in fine-grained object details. Each prompt pair is densified jointly, producing two prompts with consistent background context but distinct object-level semantics. We analyze the impact of densification in Section 5.1.3.

3.5. Image and Object Mask Generation

In this stage, we generate candidate image pairs and their corresponding object masks, which are used in the preference optimization stage to introduce object-centric guidance during training.

3.5.1. Image Generation

From each densified prompt pair $(x^{(dense)}, \tilde{x}^{(dense)})$, we generate one preferred and one non-preferred candidate image, denoted as (y_w, y_ℓ) . With N prompt pairs, N image pairs (a total of $2N$ images) are generated: $(y_{w,1}, y_{\ell,1}), (y_{w,2}, y_{\ell,2}), \dots, (y_{w,N}, y_{\ell,N})$.

3.5.2. Object Mask Generation

During image generation, we extract a binary token-level mask, referred to as the *Object Mask*, which indicates whether each visual token belongs to the object region of the image. This mask is obtained from attention weights, which makes the approach computationally efficient, as it does not require a separate segmentation model and leverages the model’s internal interactions [1, 17].

To obtain the object mask, we extract the attention distribution from a given object-representing text token over all visual tokens within the intermediate layers of the MLLM. Here, we exclude the top- k earliest and top- k latest layers to avoid unstable activations and over-smoothing. We then average them across heads and layers, and reshape the result into a 2D spatial attention map aligned with the image grid. This map is binarized into an object-specific foreground mask m using OTSU’s adaptive thresholding method [22]. We set $k = 5$ by default. By repeating this process for all objects described in the original prompt x and taking the union of the resulting masks, we obtain a single object mask M for each generated image. Detailed formulations and algorithms are provided in the Appendix.

3.6. VQA-based Preference Pair Construction

Since preference optimization is highly sensitive to data quality, OSPO includes a pair filtering and selection stage to ensure the reliability of training pairs among the N candidate image pairs. The overall process is guided by the Self-VQA results.

3.6.1. Decompositional VQA

Decompositional Visual Question Answering (VQA) is a common method for fine-grained evaluation of text–image alignment [9, 13–15]. It evaluates an image at the level of atomic semantic elements [5]. This enables fine-grained evaluation across multiple semantic dimensions. In OSPO, every prompt x in the base prompt set is decomposed by the MLLM into a set of binary Yes/No questions, $Q(x) = \{q_1, q_2, \dots, q_K\}$, where K denotes the number of atomic semantic elements. Each of the N generated image pairs is evaluated using the corresponding question set, and the MLLM outputs Yes/No probabilities for every question. The alignment score $S(y)$ for an image y is then computed as the average probability margin between the Yes and No responses: $s_k(y) = p(\text{yes} \mid y, q_k) - p(\text{no} \mid y, q_k)$, $S(y) = \frac{1}{K} \sum_{k=1}^K s_k(y)$.

3.6.2. Pair Filtering and Selection

A candidate image pair (y_w, y_ℓ) is filtered based on each image’s per-question scores $s_1(\cdot), \dots, s_K(\cdot)$ and their aggregated alignment score $S(\cdot)$ for both y_w and y_ℓ . Since the two images form a coupled pair, if either image is deemed invalid, the entire pair is discarded. In particular, the pair is discarded if the aggregated alignment score for the preferred image y_w falls below the threshold: $S(y_w) < \tau$,

where τ is a filtering threshold and we set $\tau = 0.6$ by default. Similarly, the pair is discarded if the non-preferred image y_ℓ satisfies $s_k(y_\ell) > 0, \forall k \in \{1, \dots, K\}$. By using both per-question scores s_k and the aggregated score S , we remove preference-null and preference-false pairs to avoid noisy supervision. If no candidate pair survives this process, the entire text prompt and its generated images are excluded from training.

After filtering, the final training image pair $(\hat{y}_w, \hat{y}_\ell)$ is selected as the pair whose preferred image attains the highest alignment score S . Finally, we construct **preference training triplet**: $(x, \hat{y}_w, \hat{y}_\ell)$.

3.7. Preference Optimization

Using the set of constructed preference triplets $\mathcal{D}$, we optimize the model using a combination of the Object-weighted SimPO loss and the Supervised Fine-Tuning (SFT) loss. Each loss term captures a complementary aspect of the preference learning objective.

Object-weighted SimPO Loss. Due to the nature of visual tokens, the standard SimPO objective [21], which aggregates reward over all tokens, can dilute the training signal by including many tokens that are irrelevant to the target object. Since our goal is fine-grained T2I alignment, we emphasize object-relevant visual tokens by applying spatial weights to the token-level reward in the standard SimPO. The Object-weighted SimPO loss is defined as:

$$\mathcal{L}_{\text{Obj-SimPO}} = -\mathbb{E}_{(x, y_w, y_\ell) \sim \mathcal{D}} \left[\log \sigma \left(\frac{w(y_w)}{|y_w|} \log \pi_\theta(y_w | x) - \frac{w(y_\ell)}{|y_\ell|} \log \pi_\theta(y_\ell | x) - \gamma \right) \right], \quad (1)$$

where π_θ denotes the MLLM. σ , γ , and $w(y)$ denote the sigmoid function, margin, and token-level weighting factor computed by the object mask m of y , respectively. We define $w_t = \beta \cdot (1 + \alpha m_t)$, $m_t \in \{0, 1\}$, where t denotes token position and α and β control the emphasis on object-relevant visual tokens and the overall reward scaling. This weighting scheme strengthens gradients on object-relevant tokens, producing sharper and more targeted preference signals. We set $\alpha = 1, \beta = 0.5$ by default.

SFT Loss. Visual token sequences encode spatial and structural patterns, where global coherence matters far more than the correctness of any single token. Unlike text generation, a token-level reward alone cannot effectively enforce object shape, geometry, or layout. Thus, leveraging only the PO objective may fail to provide sufficiently strong structural guidance. To complement this, we apply an SFT loss using preferred image as an anchor, ensuring consistent supervision over the full visual token sequence and reinforcing globally coherent image generation. The SFT loss is

defined as:

$$\mathcal{L}_{\text{SFT}} = -\mathbb{E}_{(x, y_w) \sim \mathcal{D}} \left[\frac{1}{|y_w|} \sum_{t=1}^{|y_w|} \log \pi_\theta((y_w)_t | x, (y_w)_{<t}) \right], \quad (2)$$

OSPO Loss. The final training objective is a weighted combination of Object-weighted SimPO loss and SFT loss: $\mathcal{L}_{\text{OSPO}} = \mathcal{L}_{\text{Obj-SimPO}} + \lambda \mathcal{L}_{\text{SFT}}$. Here, λ is a loss weight for SFT loss, which we set $\lambda = 2$ by default.

4. Experiment

4.1. Experiment Setting

We conducted experiments using the Janus-Pro-1B and Janus-Pro-7B models [3] as our primary backbones. For training, we generated a dataset consisting of 20,000 text prompts spanning four categories: Attribute, Layout, Non-spatial Relationship, and Complex Composition.

4.1.1. Benchmarks

We evaluated T2I generation capabilities on the standard benchmarks: T2I-CompBench++ [15], DPGBench [13], and GenEval [9]. Each evaluation was conducted following the standard configurations and protocols.

- **T2I-CompBench++:** Evaluates compositional image generation ability using 2,400 prompts across four categories—Attribute, Layout, Non-spatial Relationship, and Complex Composition.
- **DPGBench:** Evaluates image generation ability using 1,065 long, semantically rich prompts, which have an average token length of 83.91. It focuses on the semantic alignment of complex structures within extended narratives.
- **GenEval:** Evaluates compositional image generation ability using 550 prompts in six categories using an evaluation protocol similar to T2I-CompBench++.

4.1.2. Baselines

To evaluate the effectiveness of the OSPO framework, we compare OSPO with other self-improving preference optimization frameworks for T2I generation. Among self-improving frameworks, we include SILMM [24] and SUDER [11], both of which construct preference pairs by generating images and assigning rewards without relying on external supervision. Note that SUDER has not released its model checkpoint, thus its results are omitted for certain categories or benchmarks.

4.1.3. Implementation Details

All experiments were conducted on eight NVIDIA A100 (80GB) GPUs. Additional implementation details, including training hyperparameters, are provided in the Appendix.

Table 1. Comparison on T2I-CompBench++ [15]. ↑ indicates higher is better, with bold highlighting the best score and underline indicating the second best among the unified MLLMs.

Model	Size	Attribute ↑			Layout ↑			Non-	Complex ↑
		Color	Shape	Texture	Spatial-2D	Spatial-3D	Numeracy	Spatial ↑	
Diffusion Models									
SD-XL	–	0.6369	0.5408	0.5637	0.2032	–	–	0.3110	0.4091
DALL-E 3	–	0.7785	0.6205	0.7036	0.2865	–	–	0.3003	0.3773
FLUX.1	–	0.7407	0.5718	0.6922	0.2863	–	–	0.3127	0.3703
Unified MLLMs									
Show-o	7B	0.5600	0.4100	0.4600	0.2000	–	–	0.3000	0.2900
Unitok-MLLM	7B	0.7745	0.5195	0.6431	0.2668	0.4088	<u>0.5675</u>	0.3131	0.3617
Janus-Pro	1B	0.3542	0.2291	0.2843	0.0756	0.2367	0.2714	0.2809	0.2693
+ SILMM	1B	0.6097	0.3164	0.4474	0.1539	0.2632	0.2826	0.3122	0.3679
+ SUDER	1B	0.7765	0.5106	0.6767	0.2464	–	–	0.3130	0.3657
+ OSPO (ours)	1B	<u>0.8349</u>	0.5763	0.7274	<u>0.3077</u>	<u>0.4029</u>	0.5653	<u>0.3158</u>	<u>0.3935</u>
Janus-Pro	7B	0.5215	0.3272	0.4050	0.1654	0.2679	0.4431	0.3109	0.3868
+ SILMM	7B	0.7394	0.4325	0.5796	0.2105	0.3572	0.5073	0.3113	0.3725
+ SUDER	7B	0.7824	<u>0.5786</u>	<u>0.7292</u>	0.2524	–	–	0.3141	0.3858
+ OSPO (ours)	7B	0.8567	0.6386	0.7727	0.3562	0.3926	0.5908	0.3161	0.4147

Table 2. Comparison on DPGBench [13] and GenEval [9]. ↑ indicates higher is better, with bold indicating the best score and underline indicating the second best among the unified MLLMs.

Model	Size	DPGBench ↑						GenEval ↑						
		Attribute	Relation	Entity	Other	Global	Overall	Single	Two	Count	Colors	Position	Color Attribution	Overall
Diffusion Models														
SD-XL	-	82.43	80.91	86.76	80.41	83.27	74.65	0.98	0.74	0.39	0.85	0.15	0.15	0.55
DALL-E 3	-	89.61	88.39	90.58	89.83	90.97	83.50	0.96	0.87	0.47	0.83	0.43	0.43	0.67
SD3-Medium	-	-	-	-	-	-	-	0.99	0.94	0.72	0.89	0.33	0.60	0.74
Unified MLLMs														
Show-o2	7B	<u>89.96</u>	91.81	<u>91.78</u>	91.64	89.00	86.14	1.00	0.87	0.58	0.92	0.52	0.62	0.76
Emu3	8B	86.33	90.61	87.17	<u>89.75</u>	87.54	81.60	0.98	0.69	0.33	0.78	0.15	0.16	0.52
Unitok-MLLM	7B	88.37	91.39	88.13	87.54	83.98	81.87	<u>0.99</u>	0.75	0.39	0.80	0.21	0.41	0.59
Janus-Pro	1B	88.17	88.98	88.63	88.30	87.58	82.63	<u>0.99</u>	0.82	0.48	0.89	0.62	0.57	0.73
+ SILMM	1B	89.87	<u>91.91</u>	88.67	82.60	78.22	83.56	<u>0.99</u>	0.86	0.48	0.89	0.70	0.62	0.75
+ SUDER	1B	-	-	-	-	-	-	<u>0.99</u>	0.87	0.54	0.90	0.74	0.62	0.78
+ OSPO (ours)	1B	90.42	87.24	89.09	87.08	89.21	84.46	<u>0.99</u>	<u>0.89</u>	0.48	0.92	0.67	0.61	0.76
Janus-Pro	7B	86.11	92.72	88.30	88.85	82.62	83.81	0.98	0.87	0.57	0.89	0.78	0.67	0.80
+ SILMM	7B	89.47	91.26	90.33	88.37	91.46	84.56	0.98	0.87	0.58	0.90	0.77	<u>0.71</u>	0.80
+ SUDER	7B	-	-	-	-	-	-	<u>0.99</u>	<u>0.89</u>	0.70	0.92	<u>0.82</u>	<u>0.71</u>	0.84
+ OSPO (ours)	7B	89.03	89.45	92.14	84.22	<u>91.09</u>	<u>85.61</u>	1.00	0.90	<u>0.63</u>	<u>0.91</u>	0.83	0.72	<u>0.83</u>

4.2. Main Results

As presented in Table 1, the OSPO-trained model outperforms all self-improving baselines on T2I-CompBench++ for both the 1B and 7B model scales. Notably, substantial improvements are observed in the Attribute category, while the Non-spatial and Complex categories show relatively smaller improvements. This disparity arises from differences in the evaluation metrics across categories, particularly the CLIP-T score [10, 25], which is used for the Non-spatial and Complex categories, has limited sensitivity within its scoring range. In this context, comparisons with

other methods and models show that our model performs comparably well in those categories.

Our models of different sizes show consistent gains on other benchmarks, demonstrating the robustness of OSPO across scales. In Table 2, our 7B model achieves the second-highest overall score among unified MLLMs on DPGBench, while attaining the highest overall score among Janus-Pro-based self-improving methods. Moreover, it achieves the second-best overall score on the GenEval benchmark with only a marginal performance gap. This gap mainly comes from the Count category, a known weakness



Figure 3. Qualitative examples from Janus-Pro-7B + OSPO on T2I-CompBench++, GenEval, and DPGBench prompts.

of MLLMs—especially in generation—limiting the gains of a purely self-improving approach. In contrast, SUDER may benefit from external COCO [2] image data, allowing it to learn accurate counting signals that our self-generated framework cannot access. To wrap up, results in Table 2 demonstrate strong generalization ability of our model to out-of-distribution input prompts. Additionally, we evaluate general T2I generation performance beyond compositional ability using CLIP score. Please refer to the Appendix. Qualitative examples generated by the OSPO-trained model are shown in Figure 3, with additional samples provided in the Appendix.

5. Analysis

We conducted several experiments to analyze the contributions of individual components within the OSPO framework. All experiments were conducted using the Janus-Pro-7B model, focusing on three key aspects: (1) preference data (2) training loss and (3) computational cost. Additional analyses on iterative training, alternative 7B backbones, and data-generation biases are included in the Appendix.

5.1. Object-centric Preference Data Construction

OSPO is a self-improving framework that generates object-centric preference data that is effective for training, as shown in Figure 4 representing low ratio of Preference-Null Pairs. It generates its own training data from scratch, en-

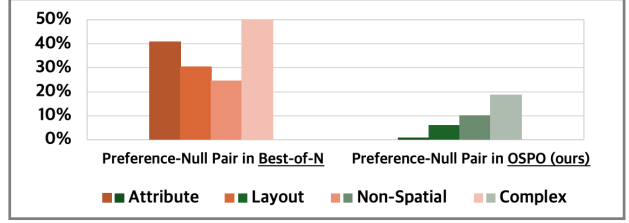


Figure 4. Category-wise comparison of *preference-null* image pairs between the (Left) Best-of-N baseline and (Right) our OSPO strategy. Lower is better.

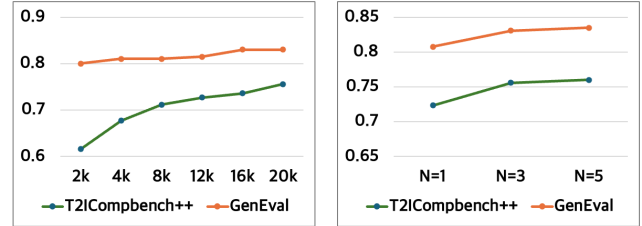


Figure 5. (Left) Effect of sample size on T2I-Compbench++ (attribute) and GenEval (overall) scores. (Right) Effect of candidate image pair size on T2I-Compbench++ (attribute) and GenEval (overall) scores. Higher is better for all benchmarks.

abling the data volume to scale with the available compute. Therefore, we first investigate two factors that influence its performance: the sample size, which refers to the total number of text prompts x (see Section 3.3), and the candidate image pair size N (see Sections 3.4 and 3.5), which refers to the number of image pairs produced per prompt.

5.1.1. Sample Size

As shown in Figure 5, model performance consistently improves as the total number of training samples increases. Notably, even with a minimal dataset, the model still achieves a substantial performance gain over the baseline. This highlights the data efficiency of OSPO framework.

5.1.2. Candidate Image Pair Size

We further analyze the effect of the candidate image pair size on overall performance. As shown in Figure 5, sampling multiple candidate pairs per prompt leads to consistent performance improvements compared to using a single pair. Although the performance gain gradually stabilizes with larger pair sizes, the results suggest that a moderate number of candidates is sufficient to capture diverse and informative training signals.

5.1.3. Image Sampling and Pair Filtering/selection

Effect of Prompt Densification on Image Fidelity. To verify that densified prompts with richer semantic descriptions really lead to more accurate text-image alignment and better model performance, we conducted a comparative study between two settings: generating images from the

Table 3. Effect of filtering and selection strategy in two different settings (with and without densification) construction stage on T2I-Compbench++ (attribute) and GenEval (overall) scores. Higher is better for all benchmarks.

Model	Filtering	Selection	T2I++↑	GenEval↑
Janus-Pro-7B			0.418	0.796
OSPO w/ Densification	✗	✗	0.716	0.813
	✓	✗	0.725	0.810
	✗	✓	0.724	0.805
	✓	✓	0.756	0.831
OSPO w/o Densification	✗	✗	0.618	0.816
	✓	✗	0.549	0.814
	✗	✓	0.627	0.814
	✓	✓	0.641	0.823

original prompt x and from its densified version $x^{(dense)}$. As shown in Table 3, training model with OSPO with densification stage consistently yields higher performance on both benchmarks. Therefore these results validate the effectiveness of the densification process. Also, the result of text-image alignment evaluation is provided in the Appendix.

Effect of Filtering and Selection Strategy. We further analyze the interaction between the filtering–selection stage and the prompt densification stage. These two stages are closely interdependent: the effectiveness of the filtering and selection processes largely depends on the diversity and semantic richness of the generated image samples. To capture this dependency, we investigate the combined impact of densification and filtering on benchmark performance, highlighting how the quality and distribution of generated samples directly influence the downstream preference pair selection process. Table 3 shows that the filtering and selection strategies consistently improve benchmark performance, regardless of whether prompt densification is applied. Notably, the effect becomes more pronounced when prompts are not densified and images are generated directly. This indicates that filtering and selection play a more critical role in mitigating noise and ensuring data quality when training on images with incomplete semantic fidelity. Additional experiments regarding the filtering threshold are provided in the Appendix.

5.2. Training Loss Design for Object-centric PO

As shown in Table 4, both loss components contribute to consistent performance gains across benchmarks. In particular, improvements are notable in spatial text–image alignment (the *Layout* category in T2I-CompBench++ and the *Position* category in GenEval). This suggests that the Object-weighted SimPO loss effectively focuses learning on object-relevant regions, while the SFT loss provides complementary structural supervision by acting as a stabilizing anchor during training. Together, these objectives en-

Table 4. Ablation study on loss components with T2I-Compbench++ (attribute, layout) and GenEval (overall, position) scores. Higher is better for all benchmarks.

Model	$\mathcal{L}_{\text{SimPO}}$	$\mathcal{L}_{\text{SimPO}}^{(w)}$	$\mathcal{L}_{\text{SFT}}$	T2I++↑		GenEval↑	
				Attr.	Layout	Over.	Position
Janus-Pro-7B				0.418	0.292	0.796	0.570
+ OSPO (ours)	✓			0.779	0.416	0.785	0.778
		✓		0.776	0.428	0.794	0.795
	✓		✓	0.755	0.437	0.827	0.795
		✓	✓	0.756	0.447	0.831	0.828

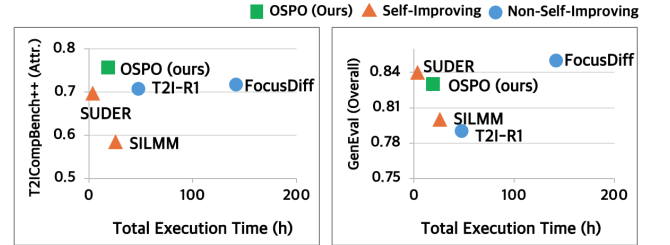


Figure 6. Performance–Efficiency trade-off across preference optimization frameworks for T2I generation. (Left) Comparison of T2I-CompBench++ (attribute) scores against execution cost. (Right) Comparison of GenEval (overall) scores against execution cost. Higher is better for all benchmarks.

hance the model’s text-faithful and spatially coherent image synthesis.

5.3. Computational Cost

Finally, we analyze the time cost of the OSPO framework compared with other preference optimization frameworks including both self-improving frameworks [11, 24] and non-self-improving frameworks [16, 23]. The time cost was measured on a system equipped with eight NVIDIA A100 (80 GB) GPUs for one iteration of each framework. We first compare OSPO with SILMM [24]. As shown in Figure 6 OSPO achieves higher benchmark performance while being more time-efficient, primarily because it generates smaller, targeted candidate sets by decoupling the source prompt. We also compare OSPO with T2I-R1 [16] and FocusDiff [23]. T2I-R1 trains the model with GRPO [28] using multiple reward models, and FocusDiff combines SFT and GRPO with training data from image-editing datasets to target fine-grained alignment. As shown in Figure 6, OSPO achieves comparable performance to these methods but with substantially lower computational cost, demonstrating its efficiency.

6. Conclusion

We propose an **Object-centric Self-improving Preference Optimization (OSPO)** framework for T2I generation with

MLLMs. OSPO leverages the unified multimodal capabilities of MLLMs to self-enhance fine-grained text–image alignment. By self-generating object-centric preference data that emphasize fine-grained semantic differences and applying object-weighted preference optimization, the model effectively self-improves its ability to capture detailed object-level semantics. Experiments on compositional image generation benchmarks show that OSPO greatly improves fine-grained alignment in T2I generation.

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